

Gleb Stachev

Embedded Systems & Full Stack Developer

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SUMMARY

Results-driven Engineer with expertise in developing autonomous systems, IoT devices, and embedded software (Python, C++, JS). Proven track record of taking complex hardware projects from concept to functional prototype, including PCB design, CAD enclosure modeling, and high-performance backend development for real-time data processing. Specialized in radio-frequency (RF) detection systems and UAV flight platforms.

TECHNICAL SKILLS

Languages: Python, C++, JavaScript.

Hardware & IoT: ESP32, Raspberry Pi Pico, UART, SPI, I2C, PWM, I2S (Audio Streaming).

Engineering: PCB Design (DipTrace), CAD Modeling (FreeCAD), 3D Printing (PETG/PLA prototyping).

Software & Computer Vision: Flask, PostgreSQL, SQL, Firebase, Azure, Google SSO, REST API, WebSockets (Real-time).

TECHNICAL PROJECTS

Cerebro: Analog Video Scanning System | 2023 – Present

- Developed a specialized hardware detector based on **Raspberry Pi Pico** for real-time interception and display of analog video feeds.
- Engineered the system to bypass false RF triggers, focusing exclusively on valid analog video signals across **1.2 GHz, 2.4 GHz, 3.3 GHz, and 5.8 GHz** bands.
- Designed a custom **PCB** and a ruggedized field enclosure using **FreeCAD** and 3D printing (PETG).
- Implemented I2C communication protocols to transmit detected frequency data.

High-Endurance Multi-Device Audio Streaming System | 2026 – Present

- Engineered a scalable **multi-device architecture** based on **ESP32**, capable of simultaneous real-time audio streaming from multiple devices.
- Optimized firmware for **I2S** digital microphones and **WebSockets**, ensuring stable 16-hour continuous data flow with minimal power consumption per device.
- Developed a robust backend using **Python** and **FastAPI** to handle concurrent high-bandwidth audio streams and synchronized data logging.
- Integrated a centralized management interface to monitor device health and frequency data across the entire network.

Heavy-Lift Long-Endurance FPV Platform | 2024 – Present

- Engineered and calibrated a heavy-lift FPV drone platform designed for **extended flight times** and specialized payloads.
- Optimized power distribution and motor efficiency to maximize battery life for long-range reconnaissance.
- Managed complex system integration, including flight controller tuning and high-gain antenna systems for stable signal transmission.

SmartCrosshair (Pet Project)

- Implemented a system of interchangeable crosshair styles that automatically or manually adjust colors to maintain visibility and target acquisition across diverse terrain backgrounds.

EXPERIENCE

Corporal National Guard of Ukraine	Ukraine 05/2015 - 05/2016
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- K9 unit.

Freelance Embedded & RF Engineer Self-Employed	Eilat - Tel Aviv, Israel 2023 – Present
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- **Cerebro (R&D):** Designed an analog video reconnaissance system, managing the full lifecycle from custom PCB design to ruggedized field enclosure fabrication.
- **FPV drone platform:** engineering and calibrating, 3d parts creating.

Embedded Engineer JUPI plus (remote) jupiplus.com	Warsaw, Poland 06/2023 - present
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- **Audio Streaming System:** Engineered the ESP32-based multi-device architecture for high-endurance real-time streaming.
- **Educational Web Portal:** Developed a custom portal for a U.S.-based organization, streamlining management for thousands of users and educational chapters (Implemented scalable SSR architectures and RESTful APIs to handle complex administrative workflows.).

EDUCATION

Full Stack Web Development, Python and JavaScript program Developers Institute, TLV Coding Bootcamp	Tel Aviv, Israel 11/2022 - 05/2023
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Master's degree Vinnytsa Institute of Trade and Economics of State University of Trade and Economics	Vinnytsa, Ukraine 06/2009- 06/2014
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LANGUAGES

Language - level (**Ukrainian, Russian** - native speaker, **Hebrew** - fluent, **English** - intermediate)